

Supplementary Material: Supervise, Intervene, Improve

Anonymous Author(s)

S1 Scope and Numerical Provenance

This supplement accompanies three independent within-participant studies. It includes the retrospective common-recovery analysis and the retained original analyses: Supervision $n = 24$, Control $n = 24$, and Fleet $n = 24$. Control has 143 of 144 planned task blocks, with one FACTR–T-Shape Assembly block unavailable. All 24 Control participants remain included; all condition questionnaires and complete rankings are present. A completed offline policy-development comparison appears in Section S8; independent final-test evaluation remains outstanding. The original Supervision immediate-only results are identified as secondary; current common-recovery estimates, contrasts, and sensitivity analyses appear in Section S12. Studies are not pooled.

The tables below report the outcome-specific estimates, comparisons, and analysis definitions. The main-paper estimates average Cup Insertion and T-Shape Assembly equally while task and retained interactions remain in the A/B objective models. Subjective ratings summarize conditions, not separate task observations. Ratios are in the listed numerator/denominator direction. Pairwise confidence intervals are unadjusted; significance requires the supplied Holm-adjusted p below .05.

S2 Subjective Ratings and Rankings

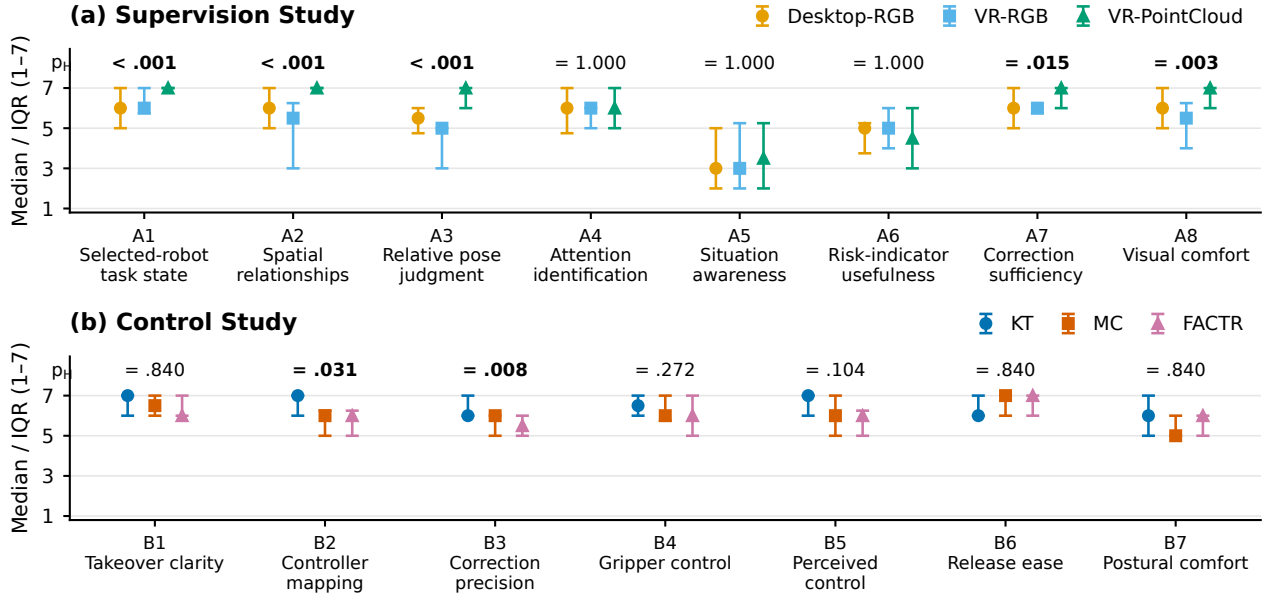


Figure S1: Complete agreement-item profiles for Supervision ($n = 24$) and Control ($n = 24$). Points and bars are condition medians and IQRs on a 1–7 scale; higher indicates greater agreement. Numbers above each item are item-family Holm-adjusted Friedman p_H values, with bold indicating $p_H < .05$. IQRs are descriptive spread, not confidence intervals, and the omnibus tests do not identify significant pairs. Exact wording, Kendall’s W , and all follow-ups are in the tables below. These panels represent independent cohorts.

Main Figure 8 displays all six NASA-TLX dimensions. The full TLX pairwise tables below retain every dimension, including null results. In Fleet, the corrected differences concern increased Effort at F6/F9 versus F1 and increased

Temporal Demand at F9 versus F1/F3. The other workload dimensions did not show corrected differences. All displayed TLX estimates and differences use 0–20 units. Validated model exports on 0–100 are divided by five, including CI endpoints; p-values and standardized effects are unchanged. Formal A/B database aliases agree exactly under this conversion. Formal Fleet raw responses also use 0–20 despite a legacy database label. Using the administered anchors confirmed by the authors, Performance is 0=Failure and 20=Perfect (higher is better); the other five dimensions are lower-is-better. Scores are not reverse-coded. Source composite entries remain in the evidence files but are omitted from the displayed tables because they average opposite scale directions.

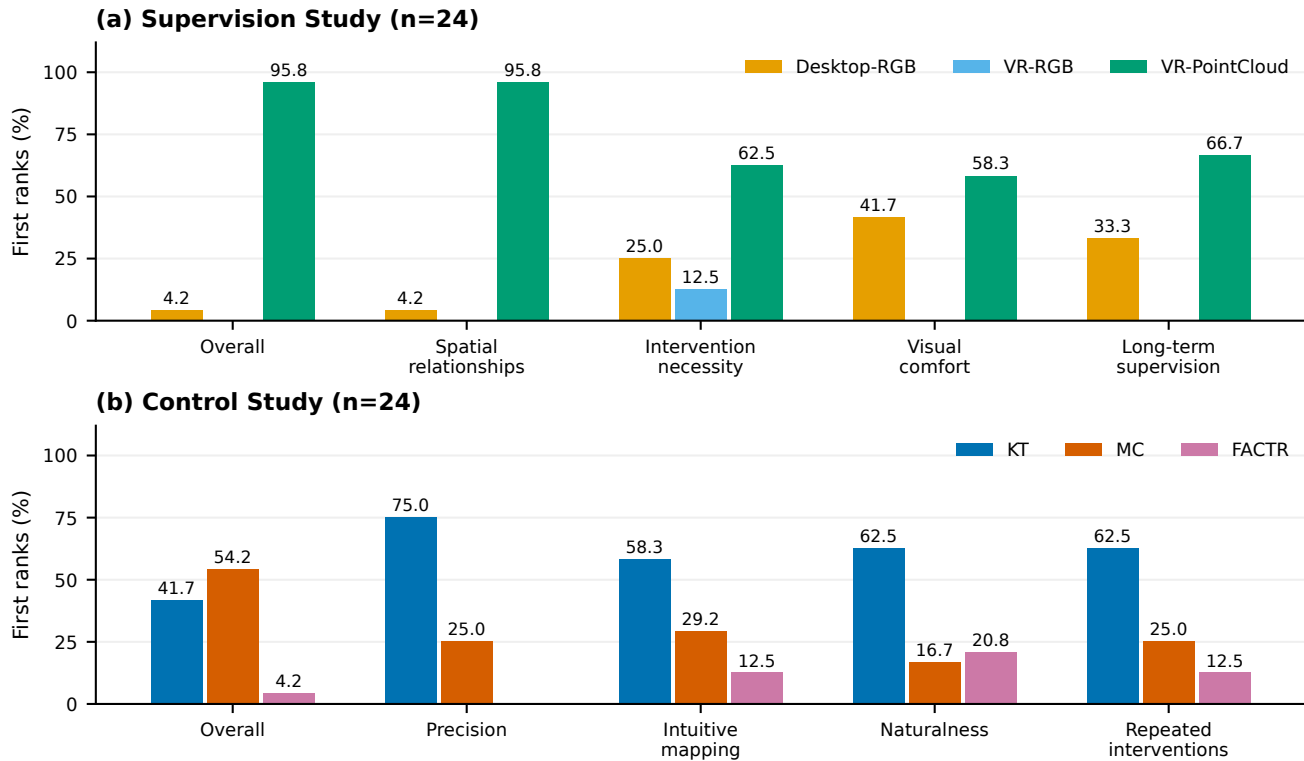


Figure S2: First-ranked conditions from complete rankings, with percentages above bars (rounded to one decimal). Supervision has 24 respondents per question and Control 24. Percentages divide each condition’s first-rank count by its cohort denominator. These descriptive shares derive from complete rankings; they are not additional forced-choice observations or objective performance. No uncertainty bars or significance symbols are implied. Complete-rank tests are reported separately below.

S3 Participant Observations and Outcome Definitions

The following source figures retain the original participant observations; the original Supervision immediate-only endpoint is shown for comparison with the harmonized analysis in Section S12. Vector condition colors in the A/B outcome figures have been aligned with the main paper; data geometry and numerical values have not been changed. The supplied aggregate CSVs do not permit recreating the participant observations independently, so their provenance is the source analysis figures.

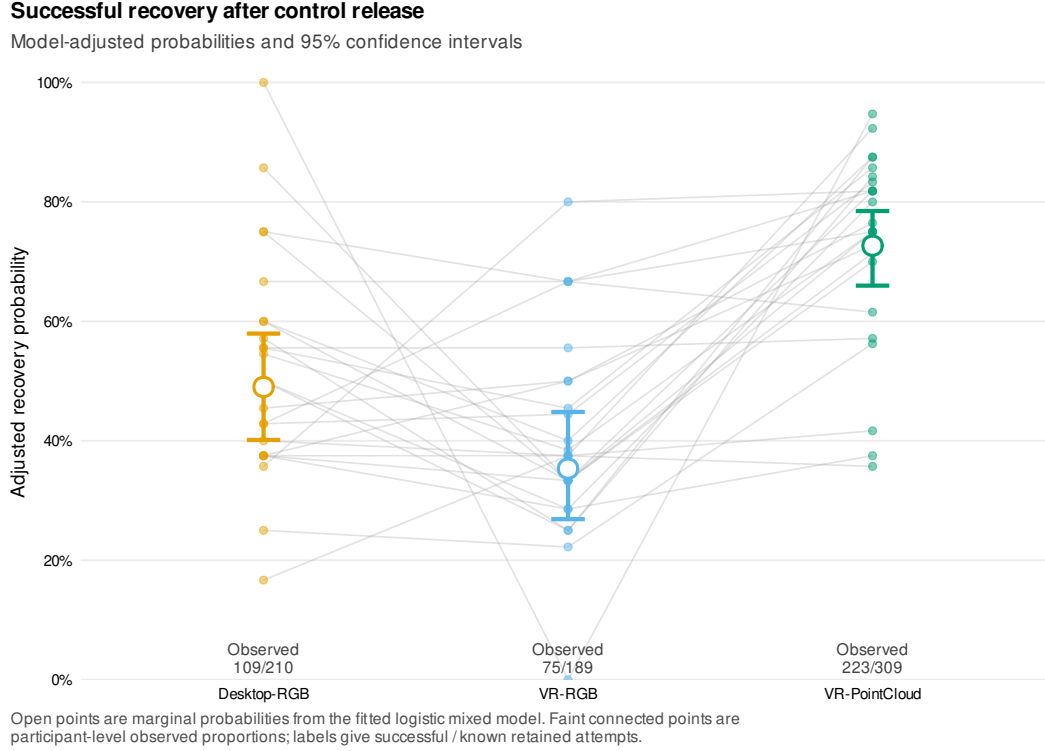


Figure S3: Original Supervision immediate-only recovery (secondary): 24 participants, 708 retained attempts. Small linked observations are participant summaries; large estimates and 95% CIs come from the task-adjusted logistic mixed model, equally averaged over both tasks. Higher recovery is favorable. Holm-adjusted contrasts are in Section S6.1; this figure does not depict recovery during human control.

S4 Analysis Details and Sensitivity Boundaries

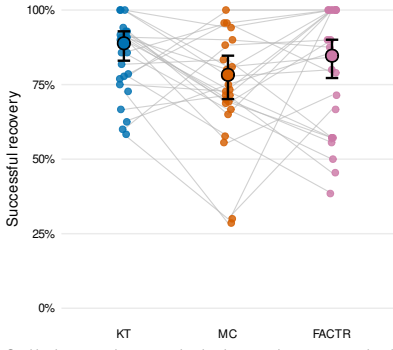
S4.1 Repeated Measures and Multiplicity

The A/B recovery and time models retain condition, task, condition order, and retained interactions. Participant and physical-scenario dependence follow the supplied model-simplification files. Supervision risk/preparation use a fixed participant block after a singular participant random intercept; the Supervision repeated-intervention model removes its sparse interaction and scenario effect, while Control retains its full candidate. The endpoint audit verifies repeated physical scenarios for the common recovery models; their construction is specified in Section S12. Other models retain their originally reported specifications.

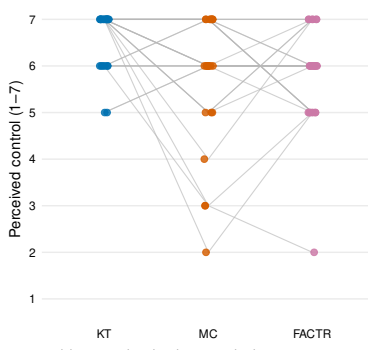
A/B objective contrasts are Holm-adjusted within each outcome's three pairs. Each study's six TLX dimensions form an 18-pair family. Supervision ordinal follow-ups form a 24-pair family; Control ordinal follow-ups a 21-pair family. Complete-ranking follow-ups form 15-pair families. Difference contrasts are printed as subtraction even when a source Fleet label used a slash; their direction remains unchanged; TLX differences alone are rescaled to 0–20 units. Item-family omnibus tests and ranking-family omnibus tests are separately corrected. Control first-rank multinomial tests use the supplied five-question Holm family, whereas the Supervision first-rank table supplies raw exact tests; the main paper reports those first ranks descriptively. Missing pairwise ordinal CIs are marked NR. Item A7 concerns confidence that the correction was sufficient before returning to autonomy; it is not confidence in deciding when to request intervention. The corrected Control precision follow-ups for KT versus MC and FACTR survive the supplied

Control Study: corrective-control interface outcomes

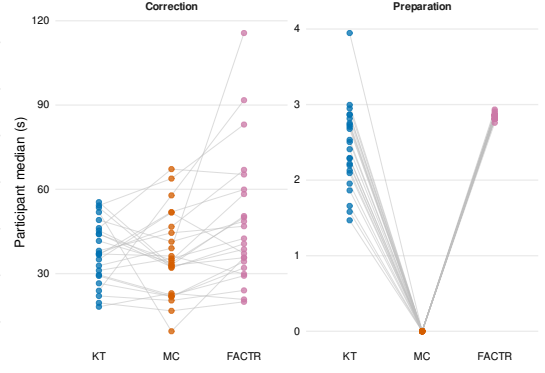
A. Successful recovery



B. Perceived control



C. Preparation and correction time



Combined outcome = human-completed or documented autonomous-continuation success; right-censored continuations are omitted.

Figure S4: Control source panels: 24 participants, 910 retained attempts, and 857 known combined-recovery outcomes. The endpoint combines immediate success with observed successful autonomous continuation; large points/bars show task-adjusted logistic estimates and 95% CIs equally averaged over both tasks. Linked points show repeated participant summaries, perceived-control ratings, and timing medians; the timing panels do not display model CIs. Higher recovery/control ratings are favorable; shorter times indicate less elapsed time. MC preparation is software-path latency.

Holm correction; Table S7 retains their rank-biserial effects and exact adjusted p-values, with NR intervals. These are perceived-precision ratings, not objective correction error.

Fleet objective comparisons use the outcome families stored in its CSV; TLX uses 36 contrasts across the six dimensions, and Fleet ratings use their separate applicable-pair family. F1 attention-switching/prioritization values are structurally absent. Fleet first-rank preferences are not combined with either A/B rankings or objective throughput. All p-values shown as less than .001 may include numerical underflow zeros in source exports; zeros are not reported as literal probabilities.

S4.2 Eligibility and Censoring

Supervision retained 708 of 740 intervention requests after excluding 25 attempts lasting 1.5 s or less and seven incomplete or invalid attempts. Control retained 910 of 972 requests after excluding 36 short and 26 incomplete or invalid attempts. These exclusions apply to attempts, with all 24 participants in each cohort retained.

Supervision logged 6771 episode starts, 5481 completions, and 1290 episodes censored at the analysis boundary; Control logged 6921, 5635, and 1286, respectively. These episode counts are not the recovery denominators. The original Supervision immediate-only endpoint uses 708 retained attempts. The harmonized primary models instead use 631 known Supervision and 857 known Control outcomes, leaving 77 and 53 unresolved attempts missing, respectively. Short-attempt filtering is applied to attempts, not participant eligibility or every outcome of an entire episode. Control has one unavailable FACTR–T-Shape Assembly block. All 24 participants contribute available task blocks and complete questionnaires; the missing cell is not imputed.

Fleet primary blocks contain 892 intervention requests and 48 attempts excluded by the ≤ 1.5 -s rule. Phase and horizon eligibility further determine the timing and recovery denominators. Fleet primary correction duration uses 772 valid active interventions and immediate recovery uses 737 evaluable checks. Coverage instead concerns 2663 effective need events, including unresolved conditions at the end. A request without an effective preceding need is not retrospectively assigned a need event. Fleet targeted 0.4 scheduled incidents per robot-minute (Section S12). Supervision and Control maintained two to three OOD cases in the nine-cell overview.

Secondary intervention frequency uses an autonomous-supervision-equivalent minute offset, distinct from wall-clock intervention throughput. Risk at request uses a tolerance-matched same-episode ACC sample in Supervision and a valid exact-policy-step sample in Control. These are selected-state descriptions, not failure-detection sensitivity or specificity. The Fleet missed-event model conditions on resolved intercepted-or-missed needs and omits F1 because it has no failure/timeout-before-intervention events. Utilization is the union of active-control intervals; concurrent demand is the time-weighted number of outstanding effective needs.

S4.3 Sensitivity and Interpretation

The A/B log-time residual diagnostics reject normality for several models. Participant-summary Friedman checks support the main Supervision recovery/correction pattern but are less conclusive for preparation. Control active correction is supported by the participant-summary test ($p_H = .012$), while combined recovery remains inconclusive ($p_H = .292$). Preparation is also supported. These analyses differ in aggregation and weighting, which qualifies interpretation of the primary finding.

The supplied Fleet sensitivity table reports stable useful-throughput direction and the F6/F9 descriptive plateau when adding the short block, using full recorded intervals, omitting order, and leaving out individual participants. Including evaluable short attempts changes the recovery denominator but does not produce a detectable fleet effect. A participant-summary throughput Friedman test is more sensitive than the primary mixed-model omnibus; it does not replace the primary Holm-adjusted pairwise inference. The Cox proportional-hazards global test was non-significant while the fleet term was borderline in the report; response timing is therefore interpreted with coverage and completed-response delay. Concurrency associations are exploratory and do not establish causal mediation.

Section S12 reports the common-endpoint refits retaining short attempts. Original participant-summary sensitivities refer to their original estimands; the Control omnibus recovery value is not a KT-MC pairwise test. Missing ordinal-effect confidence intervals remain unreported.

S4.4 Task-specific and Personalization Analyses

Task-by-condition interactions are secondary. Exploratory task-specific results are reported in Sections S12 and S14. These exploratory results complement the primary equal-weight task marginals. Participant-specific best-controller assignments are uncertain; no stable personalized-controller recommendation is inferred. Fleet free text was not formally coded and is not reported as qualitative themes.

Table S1: Task-adjusted, equal-weight marginal estimates across Cup Insertion and T-Shape Assembly. The recovery outcome includes immediate and same-episode continuation success in both studies; time estimates are geometric means in seconds. A denotes Supervision and B denotes Control. Original immediate-only results are secondary (Fig. S3; Tables S6 and S7). Intervals are model-based 95% CIs.

Study	Condition	Model	Estimate	95% CI
A	Desktop-RGB	recovery	0.684	[0.585, 0.768]
A	VR-RGB	recovery	0.560	[0.458, 0.657]
A	VR-PointCloud	recovery	0.836	[0.773, 0.884]
A	Desktop-RGB	Active correction time	43.265	[36.621, 51.114]
A	VR-RGB	Active correction time	46.385	[39.241, 54.830]
A	VR-PointCloud	Active correction time	28.910	[24.784, 33.724]
A	Desktop-RGB	Preparation delay	2.496	[2.274, 2.741]
A	VR-RGB	Preparation delay	2.846	[2.591, 3.125]
A	VR-PointCloud	Preparation delay	2.329	[2.152, 2.521]
B	KT	recovery	0.889	[0.830, 0.929]
B	MC	recovery	0.783	[0.701, 0.847]
B	FACTR	recovery	0.847	[0.772, 0.900]
B	KT	Active correction time	29.268	[25.425, 33.693]
B	MC	Active correction time	29.534	[25.708, 33.930]
B	FACTR	Active correction time	35.235	[30.487, 40.724]
B	KT	Preparation delay	2.181	[2.024, 2.351]
B	MC	Preparation delay	0.0014	[0.0013, 0.0015]
B	FACTR	Preparation delay	2.738	[2.529, 2.965]

S5 Questionnaire Wording and Complete Omnibus Tests

Table S2: Exact Supervision and Control item stems from the supplied pairwise CSVs. Agreement ratings use the verified 1–7 anchors; final comparison questions request complete rankings. Fleet wording is not reconstructed from labels.

Item	Wording
A1	I could clearly understand what the selected robot was doing and what correction was required.
A2	The task-relevant objects and their spatial relationships were easy to understand.
A3	I could accurately judge the relative position and orientation of the robot, the manipulated object, and the target.
A4	It was easy to identify which robot required my attention.
A5	I remained aware of the other robots while focusing on the selected robot.
A6	The risk indicator helped me decide when to intervene.
A7	I was confident that my correction was sufficient before returning control to autonomy.
A8	The visual display was comfortable to view throughout the condition.
A overall	Rank the three supervision interfaces from best to worst overall.
A q2	Rank the supervision interfaces from best to worst for understanding the task-relevant spatial relationships.
A q3	Rank the supervision interfaces from best to worst for deciding when intervention was necessary.
A q4	Rank the supervision interfaces from best to worst for visual comfort.
A q5	Rank the supervision interfaces from most to least preferred for supervising multiple robots over a longer period.
B1	I clearly understood when manual control had begun.
B2	The robot moved in the direction I expected.
B3	I could make precise corrections with this controller.
B4	It was easy to control the gripper.
B5	I felt in control while correcting the robot.
B6	It was easy to return control to the autonomous policy.
B7	I could maintain a comfortable posture while using the controller.
B overall	Rank the three controllers from best to worst overall.
B q2	Rank the controllers from best to worst for making precise corrections.
B q3	Rank the controllers from best to worst for an intuitive relationship between your movement and the robot's movement.
B q4	Rank the controllers from most to least natural to use.
B q5	Rank the controllers from most to least preferred for repeated corrective interventions.

Table S3: Supervision Study Friedman tests, including participant-summary sensitivities and complete rankings. W is Kendall's effect size; p_H uses the family recorded in the source.

Outcome	n	χ^2	df	W	p_H
Selected-robot task state	24	18.81	2	0.392	< .001
Spatial relationships	24	19.75	2	0.411	< .001
Relative pose judgment	24	34.80	2	0.725	< .001
Attention identification	24	0.79	2	0.017	1.000
Situation awareness	24	1.48	2	0.031	1.000
Risk-indicator usefulness	24	0.97	2	0.020	1.000
Correction sufficiency	24	11.11	2	0.231	.015
Visual comfort	24	14.86	2	0.310	.003
Post-release recovery proportion	24	21.15	2	0.441	< .001
Median risk at request	21	2.00	2	0.048	.368
Median correction duration	24	29.25	2	0.609	< .001
Median request-to-ready time	24	6.58	2	0.137	.112
Retained intervention count	24	32.46	2	0.676	< .001
Repeated-intervention proportion in completed eligible episodes	24	5.48	2	0.114	.129
Overall interface ranking	24	35.58	2	0.741	< .001
Spatial relationships	24	35.58	2	0.741	< .001

Table S5: Selected model contrasts in independent cohorts ($n = 24$ each). Contrasts divide the first condition by the second. recovery is task success after release or subsequent autonomous continuation before another request. OR: odds ratio; Ratio: time ratio. Intervals are unadjusted 95% CIs; significance requires Holm $p_H < .05$. Full recovery contrasts: Section S12; other contrasts: Tables S6–S8.

Study	Outcome	Contrast	Effect	95% CI	Holm p
Supervision	recovery	Desktop-RGB / VR-PointCloud	OR 0.423	[0.265, 0.674]	$p_H < .001$
Supervision	recovery	VR-RGB / VR-PointCloud	OR 0.249	[0.154, 0.403]	$p_H < .001$
Control	recovery	KT / MC	OR 2.212	[1.393, 3.511]	$p_H = .002$
Control	Active time	KT / FACTR	Ratio 0.831	[0.752, 0.918]	$p_H < .001$
Control	Active time	MC / FACTR	Ratio 0.838	[0.760, 0.925]	$p_H < .001$
Fleet	Coverage	F3 / F1	OR 0.051	[0.023, 0.111]	$p_H < .001$
Fleet	Throughput	F9 / F1	Rate ratio 1.316	[1.059, 1.636]	$p_H = .080$

Outcome	n	χ^2	df	W	p_H
Intervention necessity	24	11.58	2	0.241	.003
Visual comfort	24	23.58	2	0.491	< .001
Long-term multi-robot supervision	24	26.08	2	0.543	< .001

Table S4: Control Study Friedman tests, including participant-summary sensitivities and complete rankings. W is Kendall's effect size; p_H uses the family recorded in the source.

Outcome	n	χ^2	df	W	p_H
Takeover clarity	24	2.00	2	0.042	.840
Controller mapping	24	10.53	2	0.219	.031
Correction precision	24	13.58	2	0.283	.008
Gripper control	24	5.38	2	0.112	.272
Perceived control	24	7.75	2	0.162	.104
Release ease	24	0.65	2	0.014	.840
Postural comfort	24	2.55	2	0.053	.840
Combined successful-recovery proportion	24	3.85	2	0.080	.292
Median risk at request	24	7.00	2	0.146	.091
Median correction duration	24	11.58	2	0.241	.012
Median request-to-ready time	24	39.00	2	0.812	< .001
Retained intervention count	24	22.93	2	0.478	< .001
Repeated-intervention proportion in completed eligible episodes	24	2.30	2	0.048	.316
Overall controller ranking	24	5.25	2	0.109	.072
Correction precision	24	14.58	2	0.304	.003
Intuitive mapping	24	14.33	2	0.299	.003
Naturalness	24	12.25	2	0.255	.007
Repeated interventions	24	10.58	2	0.220	.010

S6 Complete Pairwise Comparisons

S6.1 Supervision Study

Table S6: Supervision Study complete supplied pairwise comparisons. Effects use the CSV scale (odds/rate/time ratios, point differences, or matched rank-biserial effects for ordinal items/rankings). NASA-TLX differences use the 0–20 scale. Intervals are unadjusted 95% CIs; p_H follows the source’s Holm family. NR means no interval supplied. OR: odds ratio; RR: rate ratio; GMR: geometric-mean ratio; HR: hazard ratio; Diff: point difference; Dlog1p/Dlogit: transformed-scale difference; r_rb: matched rank-biserial effect. A/B objective estimates are equal-weight task marginals from task-adjusted models; questionnaires summarize conditions.

Outcome/model	Contrast	Effect	95% CI	Holm p
Immediate recovery (OR)	Desktop-RGB / VR-RGB	1.759	[1.123, 2.755]	$p_H = .014$
Immediate recovery (OR)	Desktop-RGB / VR-PointCloud	0.362	[0.241, 0.543]	$p_H < .001$
Immediate recovery (OR)	VR-RGB / VR-PointCloud	0.206	[0.132, 0.320]	$p_H < .001$
Risk at request (Diff)	Desktop-RGB - VR-RGB	0.008	[-0.023, 0.040]	$p_H = .725$
Risk at request (Diff)	Desktop-RGB - VR-PointCloud	0.022	[-0.006, 0.049]	$p_H = .360$
Risk at request (Diff)	VR-RGB - VR-PointCloud	0.013	[-0.016, 0.042]	$p_H = .725$
Risk (stale-frame sensitivity) (Diff)	Desktop-RGB - VR-RGB	0.008	[-0.019, 0.035]	$p_H = .593$
Risk (stale-frame sensitivity) (Diff)	Desktop-RGB - VR-PointCloud	0.021	[-0.003, 0.045]	$p_H = .249$
Risk (stale-frame sensitivity) (Diff)	VR-RGB - VR-PointCloud	0.014	[-0.012, 0.039]	$p_H = .593$
Active correction time (GMR)	Desktop-RGB / VR-RGB	0.933	[0.814, 1.069]	$p_H = .316$
Active correction time (GMR)	Desktop-RGB / VR-PointCloud	1.497	[1.322, 1.695]	$p_H < .001$
Active correction time (GMR)	VR-RGB / VR-PointCloud	1.604	[1.412, 1.824]	$p_H < .001$
Preparation delay (GMR)	Desktop-RGB / VR-RGB	0.877	[0.792, 0.971]	$p_H = .024$
Preparation delay (GMR)	Desktop-RGB / VR-PointCloud	1.072	[0.977, 1.176]	$p_H = .143$
Preparation delay (GMR)	VR-RGB / VR-PointCloud	1.222	[1.110, 1.345]	$p_H < .001$
Intervention rate (RR)	Desktop-RGB / VR-RGB	1.147	[0.942, 1.397]	$p_H = .172$
Intervention rate (RR)	Desktop-RGB / VR-PointCloud	0.733	[0.615, 0.874]	$p_H = .001$
Intervention rate (RR)	VR-RGB / VR-PointCloud	0.639	[0.533, 0.766]	$p_H < .001$
Repeated intervention (OR)	Desktop-RGB / VR-RGB	0.430	[0.152, 1.215]	$p_H = .239$
Repeated intervention (OR)	Desktop-RGB / VR-PointCloud	0.942	[0.334, 2.653]	$p_H = .910$
Repeated intervention (OR)	VR-RGB / VR-PointCloud	2.193	[0.912, 5.275]	$p_H = .239$
TLX Effort (Diff)	Desktop-RGB - VR-RGB	-0.750	[-2.380, 0.880]	$p_H = 1.000$
TLX Effort (Diff)	Desktop-RGB - VR-PointCloud	5.125	[3.495, 6.755]	$p_H < .001$
TLX Effort (Diff)	VR-RGB - VR-PointCloud	5.875	[4.245, 7.505]	$p_H < .001$
TLX Frustration (Diff)	Desktop-RGB - VR-RGB	-2.042	[-3.978, -0.105]	$p_H = .314$
TLX Frustration (Diff)	Desktop-RGB - VR-PointCloud	5.542	[3.605, 7.478]	$p_H < .001$
TLX Frustration (Diff)	VR-RGB - VR-PointCloud	7.583	[5.647, 9.520]	$p_H < .001$
TLX Mental Demand (Diff)	Desktop-RGB - VR-RGB	-1.042	[-2.665, 0.582]	$p_H = 1.000$
TLX Mental Demand (Diff)	Desktop-RGB - VR-PointCloud	6.333	[4.710, 7.957]	$p_H < .001$
TLX Mental Demand (Diff)	VR-RGB - VR-PointCloud	7.375	[5.752, 8.998]	$p_H < .001$
TLX Performance (Diff)	Desktop-RGB - VR-RGB	-0.167	[-3.237, 2.904]	$p_H = 1.000$
TLX Performance (Diff)	Desktop-RGB - VR-PointCloud	-2.875	[-5.946, 0.196]	$p_H = .460$
TLX Performance (Diff)	VR-RGB - VR-PointCloud	-2.708	[-5.779, 0.362]	$p_H = .494$
TLX Physical Demand (Diff)	Desktop-RGB - VR-RGB	-0.458	[-1.891, 0.975]	$p_H = 1.000$
TLX Physical Demand (Diff)	Desktop-RGB - VR-PointCloud	2.375	[0.942, 3.808]	$p_H = .015$
TLX Physical Demand (Diff)	VR-RGB - VR-PointCloud	2.833	[1.400, 4.266]	$p_H = .003$
TLX Temporal Demand (Diff)	Desktop-RGB - VR-RGB	-0.958	[-2.566, 0.649]	$p_H = 1.000$
TLX Temporal Demand (Diff)	Desktop-RGB - VR-PointCloud	3.542	[1.934, 5.149]	$p_H < .001$
TLX Temporal Demand (Diff)	VR-RGB - VR-PointCloud	4.500	[2.893, 6.107]	$p_H < .001$
A1 Task state (r_rb)	Desktop-RGB - VR-RGB	0.152	NR	$p_H = 1.000$

Outcome/model	Contrast	Effect 95% CI	Holm p
A1 Task state (r_rb)	Desktop-RGB - VR-PointCloud	-1.000 NR	$p_H = .026$
A1 Task state (r_rb)	VR-RGB - VR-PointCloud	-1.000 NR	$p_H = .026$
A2 Spatial relations (r_rb)	Desktop-RGB - VR-RGB	0.431 NR	$p_H = 1.000$
A2 Spatial relations (r_rb)	Desktop-RGB - VR-PointCloud	-0.926 NR	$p_H = .020$
A2 Spatial relations (r_rb)	VR-RGB - VR-PointCloud	-0.953 NR	$p_H = .008$
A3 Relative pose (r_rb)	Desktop-RGB - VR-RGB	0.601 NR	$p_H = .414$
A3 Relative pose (r_rb)	Desktop-RGB - VR-PointCloud	-1.000 NR	$p_H = .002$
A3 Relative pose (r_rb)	VR-RGB - VR-PointCloud	-1.000 NR	$p_H < .001$
A4 Attention (r_rb)	Desktop-RGB - VR-RGB	-0.132 NR	$p_H = 1.000$
A4 Attention (r_rb)	Desktop-RGB - VR-PointCloud	-0.359 NR	$p_H = 1.000$
A4 Attention (r_rb)	VR-RGB - VR-PointCloud	-0.258 NR	$p_H = 1.000$
A5 Awareness (r_rb)	Desktop-RGB - VR-RGB	-0.073 NR	$p_H = 1.000$
A5 Awareness (r_rb)	Desktop-RGB - VR-PointCloud	-0.444 NR	$p_H = 1.000$
A5 Awareness (r_rb)	VR-RGB - VR-PointCloud	-0.136 NR	$p_H = 1.000$
A6 Risk-cue usefulness (r_rb)	Desktop-RGB - VR-RGB	-0.187 NR	$p_H = 1.000$
A6 Risk-cue usefulness (r_rb)	Desktop-RGB - VR-PointCloud	0.019 NR	$p_H = 1.000$
A6 Risk-cue usefulness (r_rb)	VR-RGB - VR-PointCloud	0.203 NR	$p_H = 1.000$
A7 Correction sufficiency (r_rb)	Desktop-RGB - VR-RGB	0.103 NR	$p_H = 1.000$
A7 Correction sufficiency (r_rb)	Desktop-RGB - VR-PointCloud	-0.846 NR	$p_H = .114$
A7 Correction sufficiency (r_rb)	VR-RGB - VR-PointCloud	-0.824 NR	$p_H = .048$
A8 Visual comfort (r_rb)	Desktop-RGB - VR-RGB	0.275 NR	$p_H = 1.000$
A8 Visual comfort (r_rb)	Desktop-RGB - VR-PointCloud	-0.717 NR	$p_H = .217$
A8 Visual comfort (r_rb)	VR-RGB - VR-PointCloud	-0.869 NR	$p_H = .028$
Overall ranking (r_rb)	Desktop-RGB - VR-RGB	-0.440 NR	$p_H = .172$
Overall ranking (r_rb)	Desktop-RGB - VR-PointCloud	0.940 NR	$p_H < .001$
Overall ranking (r_rb)	VR-RGB - VR-PointCloud	1.000 NR	$p_H < .001$
Rank: Spatial relations (r_rb)	Desktop-RGB - VR-RGB	-0.440 NR	$p_H = .172$
Rank: Spatial relations (r_rb)	Desktop-RGB - VR-PointCloud	0.940 NR	$p_H < .001$
Rank: Spatial relations (r_rb)	VR-RGB - VR-PointCloud	1.000 NR	$p_H < .001$
Rank: Intervention need (r_rb)	Desktop-RGB - VR-RGB	-0.240 NR	$p_H = .508$
Rank: Intervention need (r_rb)	Desktop-RGB - VR-PointCloud	0.523 NR	$p_H = .122$
Rank: Intervention need (r_rb)	VR-RGB - VR-PointCloud	0.707 NR	$p_H = .012$
Rank: Visual comfort (r_rb)	Desktop-RGB - VR-RGB	-0.787 NR	$p_H = .005$
Rank: Visual comfort (r_rb)	Desktop-RGB - VR-PointCloud	0.253 NR	$p_H = .508$
Rank: Visual comfort (r_rb)	VR-RGB - VR-PointCloud	0.950 NR	$p_H < .001$
Rank: Long-term use (r_rb)	Desktop-RGB - VR-RGB	-0.717 NR	$p_H = .012$
Rank: Long-term use (r_rb)	Desktop-RGB - VR-PointCloud	0.467 NR	$p_H = .172$
Rank: Long-term use (r_rb)	VR-RGB - VR-PointCloud	1.000 NR	$p_H < .001$

S6.2 Control Study

Table S7: Control Study complete supplied pairwise comparisons. Effects use the CSV scale (odds/rate/time ratios, point differences, or matched rank-biserial effects for ordinal items/rankings). NASA-TLX differences use the 0–20 scale. Intervals are unadjusted 95% CIs; p_H follows the source’s Holm family. NR means no interval supplied. OR: odds ratio; RR: rate ratio; GMR: geometric-mean ratio; HR: hazard ratio; Diff: point difference; Dlog1p/Dlogit: transformed-scale difference; r_{rb} : matched rank-biserial effect. A/B objective estimates are equal-weight task marginals from task-adjusted models; questionnaires summarize conditions.

Outcome/model	Contrast	Effect	95% CI	Holm p
Combined recovery (OR)	KT / MC	2.212	[1.393, 3.511]	$p_H = .002$
Combined recovery (OR)	KT / FACTR	1.442	[0.870, 2.389]	$p_H = .156$
Combined recovery (OR)	MC / FACTR	0.652	[0.415, 1.024]	$p_H = .127$
Immediate recovery (OR)	KT / MC	2.252	[1.507, 3.366]	$p_H < .001$
Immediate recovery (OR)	KT / FACTR	1.933	[1.263, 2.958]	$p_H = .005$
Immediate recovery (OR)	MC / FACTR	0.858	[0.586, 1.257]	$p_H = .432$
Risk at request (Diff)	KT - MC	0.010	[-0.013, 0.033]	$p_H = .403$
Risk at request (Diff)	KT - FACTR	-0.026	[-0.051, -0.002]	$p_H = .074$
Risk at request (Diff)	MC - FACTR	-0.036	[-0.061, -0.012]	$p_H = .012$
Risk (nearest-time sensitivity) (Diff)	KT - MC	0.009	[-0.016, 0.033]	$p_H = .500$
Risk (nearest-time sensitivity) (Diff)	KT - FACTR	-0.028	[-0.054, -0.001]	$p_H = .078$
Risk (nearest-time sensitivity) (Diff)	MC - FACTR	-0.036	[-0.062, -0.010]	$p_H = .018$
Active correction time (GMR)	KT / MC	0.991	[0.902, 1.089]	$p_H = .850$
Active correction time (GMR)	KT / FACTR	0.831	[0.752, 0.918]	$p_H < .001$
Active correction time (GMR)	MC / FACTR	0.838	[0.760, 0.925]	$p_H < .001$
Total intervention time (GMR)	KT / MC	1.109	[1.019, 1.208]	$p_H = .017$
Total intervention time (GMR)	KT / FACTR	0.799	[0.730, 0.874]	$p_H < .001$
Total intervention time (GMR)	MC / FACTR	0.720	[0.659, 0.787]	$p_H < .001$
Preparation delay (GMR)	KT / MC	1553.328	[1440.582, 1674.897]	$p_H < .001$
Preparation delay (GMR)	KT / FACTR	0.797	[0.736, 0.863]	$p_H < .001$
Preparation delay (GMR)	MC / FACTR	0.001	[0.000474, 0.001]	$p_H < .001$
Intervention rate (RR)	KT / MC	0.920	[0.788, 1.074]	$p_H = .290$
Intervention rate (RR)	KT / FACTR	1.202	[1.017, 1.420]	$p_H = .061$
Intervention rate (RR)	MC / FACTR	1.307	[1.110, 1.538]	$p_H = .004$
Repeated intervention (OR)	KT / MC	0.372	[0.155, 0.896]	$p_H = .074$
Repeated intervention (OR)	KT / FACTR	1.481	[0.392, 5.598]	$p_H = .563$
Repeated intervention (OR)	MC / FACTR	3.981	[1.191, 13.302]	$p_H = .074$
TLX Effort (Diff)	KT - MC	-1.708	[-3.597, 0.180]	$p_H = .953$
TLX Effort (Diff)	KT - FACTR	-1.125	[-3.014, 0.764]	$p_H = 1.000$
TLX Effort (Diff)	MC - FACTR	0.583	[-1.305, 2.472]	$p_H = 1.000$
TLX Frustration (Diff)	KT - MC	-3.625	[-5.757, -1.493]	$p_H = .021$
TLX Frustration (Diff)	KT - FACTR	-1.708	[-3.840, 0.424]	$p_H = .953$
TLX Frustration (Diff)	MC - FACTR	1.917	[-0.215, 4.049]	$p_H = .953$
TLX Mental Demand (Diff)	KT - MC	-3.292	[-4.918, -1.665]	$p_H = .003$
TLX Mental Demand (Diff)	KT - FACTR	-3.042	[-4.668, -1.415]	$p_H = .008$
TLX Mental Demand (Diff)	MC - FACTR	0.250	[-1.377, 1.877]	$p_H = 1.000$
TLX Performance (Diff)	KT - MC	2.667	[0.647, 4.687]	$p_H = .152$
TLX Performance (Diff)	KT - FACTR	1.750	[-0.270, 3.770]	$p_H = .953$
TLX Performance (Diff)	MC - FACTR	-0.917	[-2.937, 1.103]	$p_H = 1.000$
TLX Physical Demand (Diff)	KT - MC	0.917	[-1.020, 2.853]	$p_H = 1.000$
TLX Physical Demand (Diff)	KT - FACTR	1.708	[-0.228, 3.645]	$p_H = .953$
TLX Physical Demand (Diff)	MC - FACTR	0.792	[-1.145, 2.728]	$p_H = 1.000$
TLX Temporal Demand (Diff)	KT - MC	-2.625	[-4.364, -0.886]	$p_H = .059$
TLX Temporal Demand (Diff)	KT - FACTR	-1.583	[-3.323, 0.156]	$p_H = .953$
TLX Temporal Demand (Diff)	MC - FACTR	1.042	[-0.698, 2.781]	$p_H = 1.000$
B1 Takeover clarity (r_{rb})	KT - MC	0.308	NR	$p_H = 1.000$
B1 Takeover clarity (r_{rb})	KT - FACTR	0.382	NR	$p_H = 1.000$
B1 Takeover clarity (r_{rb})	MC - FACTR	0.124	NR	$p_H = 1.000$
B2 Movement mapping (r_{rb})	KT - MC	0.885	NR	$p_H = .103$
B2 Movement mapping (r_{rb})	KT - FACTR	0.829	NR	$p_H = .103$
B2 Movement mapping (r_{rb})	MC - FACTR	0.019	NR	$p_H = 1.000$
B3 Precision (r_{rb})	KT - MC	0.853	NR	$p_H = .047$
B3 Precision (r_{rb})	KT - FACTR	0.843	NR	$p_H = .037$
B3 Precision (r_{rb})	MC - FACTR	-0.076	NR	$p_H = 1.000$

Outcome/model	Contrast	Effect 95% CI	Holm p
B4 Gripper control (r_rb)	KT - MC	0.533 NR	$p_H = 1.000$
B4 Gripper control (r_rb)	KT - FACTR	0.449 NR	$p_H = 1.000$
B4 Gripper control (r_rb)	MC - FACTR	-0.038 NR	$p_H = 1.000$
B5 Perceived control (r_rb)	KT - MC	0.700 NR	$p_H = .262$
B5 Perceived control (r_rb)	KT - FACTR	0.817 NR	$p_H = .081$
B5 Perceived control (r_rb)	MC - FACTR	-0.100 NR	$p_H = 1.000$
B6 Release ease (r_rb)	KT - MC	-0.091 NR	$p_H = 1.000$
B6 Release ease (r_rb)	KT - FACTR	0.127 NR	$p_H = 1.000$
B6 Release ease (r_rb)	MC - FACTR	0.143 NR	$p_H = 1.000$
B7 Postural comfort (r_rb)	KT - MC	0.298 NR	$p_H = 1.000$
B7 Postural comfort (r_rb)	KT - FACTR	0.132 NR	$p_H = 1.000$
B7 Postural comfort (r_rb)	MC - FACTR	-0.359 NR	$p_H = 1.000$
Overall ranking (r_rb)	KT - MC	-0.160 NR	$p_H = 1.000$
Overall ranking (r_rb)	KT - FACTR	-0.600 NR	$p_H = .043$
Overall ranking (r_rb)	MC - FACTR	-0.420 NR	$p_H = .390$
Rank: Precision (r_rb)	KT - MC	-0.540 NR	$p_H = .122$
Rank: Precision (r_rb)	KT - FACTR	-0.820 NR	$p_H = .003$
Rank: Precision (r_rb)	MC - FACTR	-0.193 NR	$p_H = 1.000$
Rank: Movement mapping (r_rb)	KT - MC	-0.557 NR	$p_H = .111$
Rank: Movement mapping (r_rb)	KT - FACTR	-0.830 NR	$p_H = .003$
Rank: Movement mapping (r_rb)	MC - FACTR	-0.310 NR	$p_H = .872$
Rank: Naturalness (r_rb)	KT - MC	-0.700 NR	$p_H = .023$
Rank: Naturalness (r_rb)	KT - FACTR	-0.653 NR	$p_H = .043$
Rank: Naturalness (r_rb)	MC - FACTR	0.000 NR	$p_H = 1.000$
Rank: Repeated use (r_rb)	KT - MC	-0.560 NR	$p_H = .111$
Rank: Repeated use (r_rb)	KT - FACTR	-0.683 NR	$p_H = .025$
Rank: Repeated use (r_rb)	MC - FACTR	-0.040 NR	$p_H = 1.000$

S6.3 Fleet Study

Table S8: Fleet Study complete supplied pairwise comparisons. NASA-TLX differences use the 0–20 scale. Intervals are unadjusted 95% CIs; p_H follows the source’s Holm family. Effects use the listed contrast direction: OR, odds ratio; RR, rate ratio; GMR, geometric-mean ratio; HR, hazard ratio; Diff, point difference; Dlog1p/Dlogit, transformed-scale difference. All contrasts are within the Fleet cohort.

Outcome/model	Contrast	Effect	95% CI	Holm p
Attention switching (Diff)	F6 - F3	-0.250	[-0.767, 0.267]	$p_H = .932$
Attention switching (Diff)	F9 - F3	-0.625	[-1.360, 0.110]	$p_H = .710$
Attention switching (Diff)	F9 - F6	-0.375	[-0.892, 0.142]	$p_H = .932$
Awareness (Diff)	F3 - F1	-0.750	[-1.192, -0.308]	$p_H = .015$
Awareness (Diff)	F6 - F1	-1.167	[-1.751, -0.583]	$p_H = .002$
Awareness (Diff)	F6 - F3	-0.417	[-0.896, 0.063]	$p_H = .710$
Awareness (Diff)	F9 - F1	-1.500	[-2.282, -0.718]	$p_H = .003$
Awareness (Diff)	F9 - F3	-0.750	[-1.396, -0.104]	$p_H = .274$
Awareness (Diff)	F9 - F6	-0.333	[-0.813, 0.146]	$p_H = .932$
Ability to keep up (Diff)	F3 - F1	-0.375	[-0.921, 0.171]	$p_H = .932$
Ability to keep up (Diff)	F6 - F1	-0.917	[-1.510, -0.324]	$p_H = .034$
Ability to keep up (Diff)	F6 - F3	-0.542	[-1.099, 0.016]	$p_H = .520$
Ability to keep up (Diff)	F9 - F1	-1.083	[-1.755, -0.412]	$p_H = .025$
Ability to keep up (Diff)	F9 - F3	-0.708	[-1.325, -0.092]	$p_H = .274$
Ability to keep up (Diff)	F9 - F6	-0.167	[-0.724, 0.391]	$p_H = .932$
Prioritization (Diff)	F6 - F3	-0.292	[-0.828, 0.244]	$p_H = .932$
Prioritization (Diff)	F9 - F3	-1.333	[-2.030, -0.636]	$p_H = .003$
Prioritization (Diff)	F9 - F6	-1.042	[-1.578, -0.506]	$p_H = .003$
Perceived response time (Diff)	F3 - F1	-0.875	[-1.506, -0.244]	$p_H = .085$
Perceived response time (Diff)	F6 - F1	-1.500	[-2.131, -0.869]	$p_H < .001$
Perceived response time (Diff)	F6 - F3	-0.625	[-1.256, 0.006]	$p_H = .520$
Perceived response time (Diff)	F9 - F1	-2.500	[-3.131, -1.869]	$p_H < .001$
Perceived response time (Diff)	F9 - F3	-1.625	[-2.256, -0.994]	$p_H < .001$
Perceived response time (Diff)	F9 - F6	-1.000	[-1.631, -0.369]	$p_H = .028$
TLX Effort (Diff)	F3 - F1	1.542	[0.180, 2.903]	$p_H = .640$
TLX Effort (Diff)	F6 - F1	2.583	[1.093, 4.073]	$p_H = .024$
TLX Effort (Diff)	F6 - F3	1.042	[-0.352, 2.435]	$p_H = 1.000$
TLX Effort (Diff)	F9 - F1	3.875	[2.172, 5.578]	$p_H < .001$
TLX Effort (Diff)	F9 - F3	2.333	[0.780, 3.887]	$p_H = .099$
TLX Effort (Diff)	F9 - F6	1.292	[-0.102, 2.685]	$p_H = 1.000$
TLX Frustration (Diff)	F3 - F1	0.542	[-0.663, 1.747]	$p_H = 1.000$
TLX Frustration (Diff)	F6 - F1	1.167	[-0.171, 2.504]	$p_H = 1.000$
TLX Frustration (Diff)	F6 - F3	0.625	[-0.613, 1.863]	$p_H = 1.000$
TLX Frustration (Diff)	F9 - F1	1.833	[0.280, 3.387]	$p_H = .538$
TLX Frustration (Diff)	F9 - F3	1.292	[-0.110, 2.693]	$p_H = 1.000$
TLX Frustration (Diff)	F9 - F6	0.667	[-0.571, 1.904]	$p_H = 1.000$
TLX Mental Demand (Diff)	F3 - F1	1.167	[-0.127, 2.461]	$p_H = 1.000$
TLX Mental Demand (Diff)	F6 - F1	1.750	[0.362, 3.138]	$p_H = .364$
TLX Mental Demand (Diff)	F6 - F3	0.583	[-0.734, 1.900]	$p_H = 1.000$
TLX Mental Demand (Diff)	F9 - F1	2.375	[0.826, 3.924]	$p_H = .085$
TLX Mental Demand (Diff)	F9 - F3	1.208	[-0.227, 2.644]	$p_H = 1.000$
TLX Mental Demand (Diff)	F9 - F6	0.625	[-0.692, 1.942]	$p_H = 1.000$
TLX Performance (Diff)	F3 - F1	-0.750	[-2.081, 0.581]	$p_H = 1.000$
TLX Performance (Diff)	F6 - F1	-1.208	[-2.653, 0.236]	$p_H = 1.000$
TLX Performance (Diff)	F6 - F3	-0.458	[-1.817, 0.900]	$p_H = 1.000$
TLX Performance (Diff)	F9 - F1	-2.458	[-4.092, -0.824]	$p_H = .099$
TLX Performance (Diff)	F9 - F3	-1.708	[-3.208, -0.208]	$p_H = .640$
TLX Performance (Diff)	F9 - F6	-1.250	[-2.609, 0.109]	$p_H = 1.000$
TLX Physical Demand (Diff)	F3 - F1	0.542	[-0.383, 1.467]	$p_H = 1.000$
TLX Physical Demand (Diff)	F6 - F1	0.750	[-0.587, 2.087]	$p_H = 1.000$
TLX Physical Demand (Diff)	F6 - F3	0.208	[-0.829, 1.246]	$p_H = 1.000$
TLX Physical Demand (Diff)	F9 - F1	2.125	[0.250, 4.000]	$p_H = .640$
TLX Physical Demand (Diff)	F9 - F3	1.583	[0.075, 3.091]	$p_H = .871$
TLX Physical Demand (Diff)	F9 - F6	1.375	[0.337, 2.413]	$p_H = .264$
TLX Temporal Demand (Diff)	F3 - F1	0.458	[-0.598, 1.514]	$p_H = 1.000$
TLX Temporal Demand (Diff)	F6 - F1	1.167	[-0.123, 2.457]	$p_H = 1.000$

Outcome/model	Contrast	Effect	95% CI	Holm p
TLX Temporal Demand (Diff)	F6 - F3	0.708	[-0.408, 1.824]	$p_H = 1.000$
TLX Temporal Demand (Diff)	F9 - F1	2.792	[1.153, 4.430]	$p_H = .028$
TLX Temporal Demand (Diff)	F9 - F3	2.333	[0.936, 3.730]	$p_H = .035$
TLX Temporal Demand (Diff)	F9 - F6	1.625	[0.509, 2.741]	$p_H = .125$
Active correction time (GMR)	F3 / F1	1.025	[0.913, 1.152]	$p_H = 1.000$
Active correction time (GMR)	F6 / F1	0.941	[0.839, 1.055]	$p_H = 1.000$
Active correction time (GMR)	F6 / F3	0.917	[0.824, 1.022]	$p_H = .583$
Active correction time (GMR)	F9 / F1	1.057	[0.941, 1.187]	$p_H = 1.000$
Active correction time (GMR)	F9 / F3	1.031	[0.925, 1.149]	$p_H = 1.000$
Active correction time (GMR)	F9 / F6	1.124	[1.010, 1.249]	$p_H = .188$
Need-to-control delay (GMR)	F3 / F1	2.678	[2.218, 3.235]	$p_H < .001$
Need-to-control delay (GMR)	F6 / F1	3.331	[2.727, 4.069]	$p_H < .001$
Need-to-control delay (GMR)	F6 / F3	1.244	[1.040, 1.487]	$p_H = .051$
Need-to-control delay (GMR)	F9 / F1	3.202	[2.566, 3.996]	$p_H < .001$
Need-to-control delay (GMR)	F9 / F3	1.195	[0.983, 1.454]	$p_H = .147$
Need-to-control delay (GMR)	F9 / F6	0.961	[0.805, 1.149]	$p_H = .664$
Coverage (OR)	F3 / F1	0.051	[0.023, 0.111]	$p_H < .001$
Coverage (OR)	F6 / F1	0.014	[0.006, 0.030]	$p_H < .001$
Coverage (OR)	F6 / F3	0.270	[0.209, 0.351]	$p_H < .001$
Coverage (OR)	F9 / F1	0.008	[0.004, 0.018]	$p_H < .001$
Coverage (OR)	F9 / F3	0.164	[0.127, 0.211]	$p_H < .001$
Coverage (OR)	F9 / F6	0.605	[0.486, 0.753]	$p_H < .001$
Concurrent needs (Dlog1p)	F3 - F1	0.669	[0.633, 0.705]	$p_H < .001$
Concurrent needs (Dlog1p)	F6 - F1	1.361	[1.325, 1.397]	$p_H < .001$
Concurrent needs (Dlog1p)	F6 - F3	0.692	[0.656, 0.728]	$p_H < .001$
Concurrent needs (Dlog1p)	F9 - F1	1.711	[1.675, 1.747]	$p_H < .001$
Concurrent needs (Dlog1p)	F9 - F3	1.042	[1.006, 1.078]	$p_H < .001$
Concurrent needs (Dlog1p)	F9 - F6	0.350	[0.314, 0.386]	$p_H < .001$
Missed needs (OR)	F6 / F3	4.273	[3.232, 5.649]	$p_H < .001$
Missed needs (OR)	F9 / F3	7.343	[5.581, 9.660]	$p_H < .001$
Missed needs (OR)	F9 / F6	1.719	[1.374, 2.150]	$p_H < .001$
Active-control utilization (Dlogit)	F3 - F1	0.504	[0.370, 0.638]	$p_H < .001$
Active-control utilization (Dlogit)	F6 - F1	0.553	[0.412, 0.695]	$p_H < .001$
Active-control utilization (Dlogit)	F6 - F3	0.049	[-0.085, 0.183]	$p_H = 1.000$
Active-control utilization (Dlogit)	F9 - F1	0.555	[0.401, 0.709]	$p_H < .001$
Active-control utilization (Dlogit)	F9 - F3	0.051	[-0.092, 0.194]	$p_H = 1.000$
Active-control utilization (Dlogit)	F9 - F6	0.002	[-0.132, 0.136]	$p_H = 1.000$
Immediate recovery (OR)	F3 / F1	1.110	[0.477, 2.579]	$p_H = .880$
Immediate recovery (OR)	F6 / F1	1.858	[0.751, 4.595]	$p_H = .721$
Immediate recovery (OR)	F6 / F3	1.674	[0.691, 4.055]	$p_H = .760$
Immediate recovery (OR)	F9 / F1	2.828	[1.010, 7.916]	$p_H = .287$
Immediate recovery (OR)	F9 / F3	2.549	[0.928, 6.998]	$p_H = .347$
Immediate recovery (OR)	F9 / F6	1.522	[0.524, 4.422]	$p_H = .880$
Time to human control (HR)	F3 / F1	0.083	[0.063, 0.110]	$p_H < .001$
Time to human control (HR)	F6 / F1	0.030	[0.023, 0.039]	$p_H < .001$
Time to human control (HR)	F6 / F3	0.361	[0.324, 0.402]	$p_H < .001$
Time to human control (HR)	F9 / F1	0.020	[0.015, 0.026]	$p_H < .001$
Time to human control (HR)	F9 / F3	0.236	[0.208, 0.268]	$p_H < .001$
Time to human control (HR)	F9 / F6	0.654	[0.605, 0.708]	$p_H < .001$
Useful throughput (RR)	F3 / F1	1.185	[0.949, 1.481]	$p_H = .538$
Useful throughput (RR)	F6 / F1	1.285	[1.033, 1.599]	$p_H = .122$
Useful throughput (RR)	F6 / F3	1.084	[0.882, 1.333]	$p_H = .950$
Useful throughput (RR)	F9 / F1	1.316	[1.059, 1.636]	$p_H = .080$
Useful throughput (RR)	F9 / F3	1.111	[0.904, 1.364]	$p_H = .950$
Useful throughput (RR)	F9 / F6	1.024	[0.838, 1.252]	$p_H = .950$

S7 Interface and Device Images

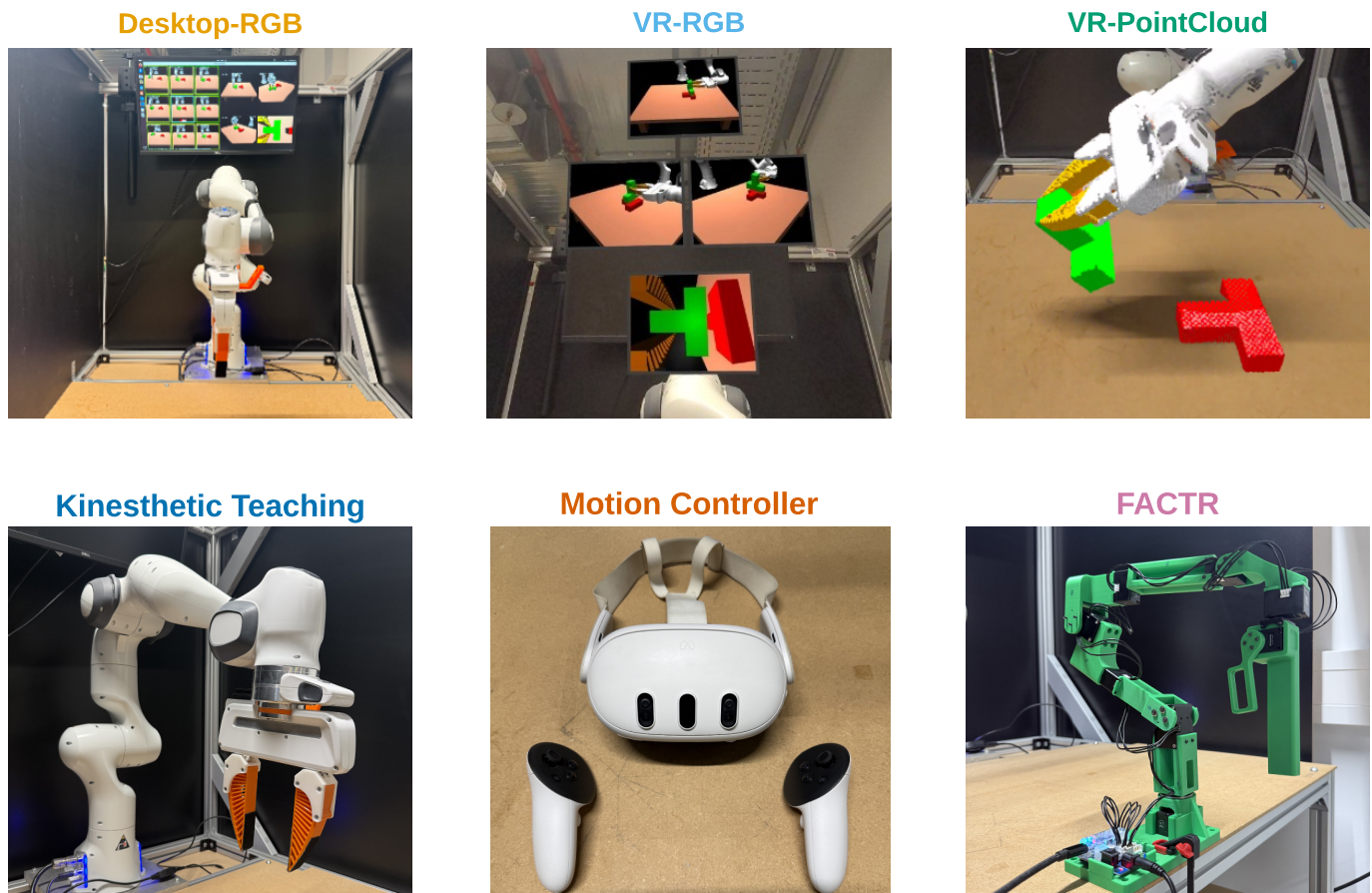


Figure S5: Actual supervision-interface and corrective-control apparatus images. The top row shows Desktop-RGB, VR-RGB, and VR-PointCloud. The bottom row shows the physical Panda, Meta Quest headset and controllers, and FACTR. These images explain the apparatus rather than a quantitative result.

S8 Offline Policy Development from Control Corrections

This section documents the matched-frame Control-study policy-development comparison. Its outcome is autonomous task success, distinct from attempt-level human-intervention recovery. Main Figure 10 reports both tasks; the tables below provide the numerical results and correction-data budgets. Additional-autonomous-data and combined-data ablations were not evaluated.

S8.1 Evaluation data sources

Reported success counts follow the author-confirmed evaluation sheet. Four entries differ from the earlier exported summaries (Table S9); corresponding trial-level records were not supplied for this revision. The retained FACTR/Cups seed-2 checkpoint at update 5,000 has 42 ID successes, whereas earlier exports list 43 at updates 1,000, 2,000, and 3,000. The highest-ID selection rule therefore remains unverified for this entry. The supplied checkpoint identities are retained, and the original exports and reported counts are stored separately.

S8.2 Data budgets and training

Only curated successful OOD intervention episodes from the Control study under VR-PointCloud were used. Each task/interface arm retained 6,677 active correction frames; preparation was excluded. KT, MC, and FACTR used identical frame budgets and 90% base / 10% intervention sampling. Matching frames does not match source-episode counts or scenario difficulty (Table S11). No extra autonomous study replay was added.

Table S9: Differences between the earlier export and the reported counts, each over 50 trials. Run/update identifies the selected model.

Task	Arm / split	Seed	Run/update	Earlier export	Reported
Cups	FACTR / ID	1	402/3000	44	43
Cups	FACTR / ID	2	410/5000	44	42
T-Shape	MC / OOD	1	427/5000	19	20
T-Shape	MC / OOD	2	435/2000	19	17

Cup Insertion initializes ACT from the 400,000-update BASE checkpoint, with learning rate 10^{-5} (backbone 10^{-6}) and chunk/execution lengths 16/16. T-Shape initializes a Diffusion policy from the 600,000-update BASE checkpoint, with learning rate 3×10^{-7} and prediction/execution lengths 8/4. Both use batch size 16, a fresh optimizer, and 5,000 fine-tuning updates. The reported comparison uses training seeds 0–2 consistently across tasks and interfaces. These are offline training configurations; the user-study policies remained fixed. The supplied package identifies architectures but does not independently document their rationale or the original training-data splits.

S8.3 Selection, evaluation, and variability

All 90 task/interface/seed/checkpoint combinations for the reported training seeds have complete 50-trial ID and 50-trial OOD validation scores. In the archived selection, the checkpoint for each training seed at 1,000, 2,000, 3,000, 4,000, or 5,000 updates is selected by highest ID successes, then highest OOD successes, then earliest update. The archived rule was applied to every arm; the reported checkpoint identities and source differences are documented in Section S8.1. The main comparison reports the original training seeds 0–2 consistently across interfaces and tasks. Both selection and reporting use these scores, so they are development estimates rather than independent held-out tests. Trials repeat a fixed scene panel; 50 trials do not represent 50 independent scenes. The supplied exports do not establish source-disjoint evaluation relative to policy-training data.

For each split, each training seed contributes its success percentage over 50 trials. For the main comparison, we report the arithmetic mean and sample standard deviation ($n - 1$ denominator) across the original three training seeds (0–2). The historical BASE reference is one trained checkpoint per task evaluated with three evaluation seeds (1–3), also 50 trials per split. Its SD measures evaluation randomness, whereas fine-tuned SD summarizes independently trained runs together with evaluation variability. SDs are not confidence intervals, and no inferential superiority test is claimed.

T-Shape uses an 80-s evaluation limit throughout. Cups retains archived 90- and 300-s limits, with BASE at 300 s. All selected Cups runs at seeds 0–1 use 300 s. At seed 2, KT uses 300 s while MC and FACTR use 90 s. Thus the Cups panel is an archive comparison, not a uniformly timed estimate of improvement. These limits define evaluation opportunity; they are not task-completion-time outcomes. Archived exports are unchanged; the four amendments above apply only to the separately identified selected-model summaries.

Table S10: Policy success: mean $\pm$ sample SD (percentage points), plus total counts across seeds for verification. BASE has three evaluation seeds; other arms use the original three training seeds (0–2). Cups has mixed evaluation limits.

Task	Arm	ID (%)	OOD (%)	ID count	OOD count
Cups	BASE	76.7 ± 9.5	12.0 ± 2.0	115/150	18/150
Cups	KT	88.0 ± 7.2	22.7 ± 5.0	132/150	34/150
Cups	MC	84.7 ± 1.2	30.0 ± 4.0	127/150	45/150
Cups	FACTR	86.0 ± 2.0	25.3 ± 4.2	129/150	38/150
T-Shape	BASE	74.7 ± 7.6	35.3 ± 1.2	112/150	53/150
T-Shape	KT	84.0 ± 4.0	40.7 ± 2.3	126/150	61/150
T-Shape	MC	82.7 ± 7.0	37.3 ± 3.1	124/150	56/150
T-Shape	FACTR	84.7 ± 5.8	39.3 ± 3.1	127/150	59/150

S8.4 Data provenance

The accompanying `learning_assets/data/` files retain all selected counts, checkpoint counts, training budgets, and source hashes. The plotting script checks the archived checkpoint rule, applies the documented count updates, and computes the reported means and sample SDs. These are arithmetic consistency checks, not independent trial rescoring.

Table S11: Curated correction budgets per training arm. All source episodes were selected as successful, and all arms use 6,677 OOD frames. Equal frames do not imply equal episodes or scene composition.

Task	Arm	Frames	Segments	Source episodes
Cups	KT	6677	15	15
Cups	MC	6677	16	16
Cups	FACTR	6677	16	16
T-Shape	KT	6677	20	20
T-Shape	MC	6677	26	26
T-Shape	FACTR	6677	24	24

S9 Participant Background and Recorded Procedure

Table S12: Age summaries from the formal questionnaire cohorts; each contains 24 background records. SD is between-participant standard deviation, not uncertainty in the mean.

Study	<i>n</i>	Mean (SD), years	Range, years
Supervision	24	23.4 (4.2)	18–39
Control	24	23.9 (3.2)	18–30
Fleet	24	24.9 (2.4)	20–30

The questionnaire database contains background and final comparison records for all 24 formal participants in each study. Additional database records are outside the included cohorts; their recruitment status must not be inferred from identifier values alone. A workflow completion timestamp is absent for one Control record despite complete questionnaires; this is not treated as participant exclusion. Participants were recruited through university advertisements and campus flyers. Participants were offered EUR 15 or study credit for sessions lasting approximately 90 min. All three studies received approval from the institutional ethics committee and followed the same written informed-consent procedure. Identifying institutional details are withheld for anonymous review. These counts describe the analyzed cohorts; additional database records are not used to infer recruitment or attrition.

Within each Supervision/Control cohort, the six condition-order assignments each occur four times. Recorded task-order assignments divide equally: twelve T-Shape Assembly → Cup Insertion and twelve Cup Insertion → T-Shape Assembly. The task-order group is associated with condition order; these assignment records do not replace actual event chronology. Fleet uses all 24 order permutations once. The database records successful practice for the formal Supervision/Control cohorts. Supervision and Control prescribed two nominal five-minute task blocks per interface. All three cohorts completed comprehension checks on the ACC cue, robot states, takeover, release, and safety, followed by interface practice and at least one successful intervention before measured interaction.

The aggregate exports contain demographics, assignment counts, and full-rank counts; participant-level free text and the questionnaire database are not redistributed. First-rank totals independently match the supplied final analysis tables. Main Figure 9 extends those summaries to all three overall ranks and separates the Fleet preference questions.

S10 Intervention Outcome Accounting and Model Specifications

Table S13: Control recovery accounting by controller. Counts are attempts; Immediate and Continuation count successes. Bounds assign every unresolved continuation to failure (lower) or success (upper); they are unadjusted descriptive bounds, not confidence intervals or a refitted sensitivity model.

Controller	Retained	Known	Immediate	Continuation	Unresolved	Bounds (%)
KT	309	299	243	10	10	81.9–85.1
MC	347	324	232	13	23	70.6–77.2
FACTR	254	234	177	12	20	74.4–82.3

Immediate success is observed at the first valid check after release. Combined recovery additionally includes successful same-episode autonomous continuation before another request. The first measures completion at hand-back; the second measures whether a correction completes the task or restores autonomy sufficiently to finish. The main paper uses one endpoint, recovery, in both interface studies. This is the retrospectively harmonized endpoint previously called combined recovery in the analysis exports. “Immediate recovery” is a component analysis reported only in this supplement. The original immediate-only Supervision model remains secondary. Independent cohorts are still analyzed separately.

The unresolved Control counts differ across controllers. Table S13 shows exactly how much each unadjusted rate could change under extreme assignments of those outcomes. These bounds do not adjust for participant, task, or order, and are not substitutes for a repeated-measures sensitivity model. They neither prove missingness is harmless nor establish the adjusted controller ordering under informative censoring. The immediate-recovery model, which uses every retained attempt, and its complete contrasts are reported in Table S7. The event audit found no recoverable later terminal outcome for the 53 unresolved Control attempts. Model-based extreme assignments and short-attempt sensitivities are reported in Section S12.

Table S14: Reported A/B recovery and timing model specifications. F denotes condition, task, their interaction, and categorical condition position; u_p and u_s are participant and scenario random intercepts. P_p is a fixed participant block. Sample sizes are model observations, not participants.

Study	Outcome	n	Predictor and response scale
Supervision	recovery	631	logit; $F + u_p + u_s$
Supervision	Immediate recovery	708	logit; $F + u_p + u_s$
Supervision	Active correction	708	log time; $F + u_p + u_s$
Supervision	Preparation	708	log time; $F + P_p + u_s$
Control	Combined recovery	857	logit; $F + u_p + u_s$
Control	Immediate recovery	910	logit; $F + u_p + u_s$
Control	Active correction	910	log time; $F + u_p + u_s$
Control	Preparation	910	log time; $F + u_p + u_s$
Control	Total intervention	910	log time; $F + u_p + u_s$

Table S14 expresses the reported A/B specifications in common notation: $F = \beta_0 + \beta_{\text{condition}} + \beta_{\text{task}} + \beta_{\text{condition} \times \text{task}} + \beta_{\text{position}}$. The recovery specifications reflect the common-endpoint refits. Their repeated physical-scenario identifier is defined in Section S12; other specifications follow the original analysis tables. The Supervision preparation model retains participant as a fixed block after a singular random-intercept fit.

Each A/B TLX dimension uses condition and categorical condition position with a participant random intercept; tasks are not separate questionnaire observations. Fleet throughput uses a log-link Poisson model with log condition-minutes as an exposure offset. Coverage and immediate recovery use logistic models. Those Fleet models include condition position and a participant random intercept after the reported slope simplifications. The log-time delay and active-correction models retain a participant intercept and a linear fleet-size slope; censored time-to-control uses participant-clustered Cox inference. The common-recovery model manifests accompany the endpoint audit.

S11 Procedure Overview and Analysis Software

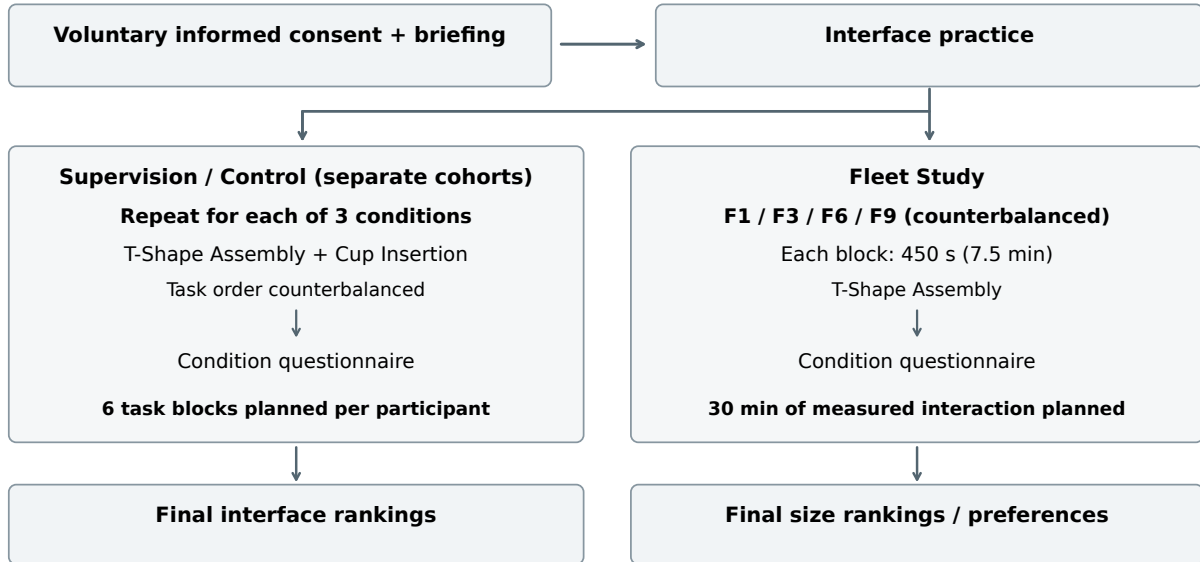


Figure S6: Procedure for the three independent cohorts ($n = 24$ each). Supervision and Control each plan three interface conditions with both tasks, giving six task blocks per participant. Fleet plans four 450-s blocks (30 min of measured interaction), followed by final preference questions. Questionnaire and practice time are additional. Supervision and Control prescribe 10 min per interface, split equally between tasks. No timing distribution or uncertainty estimate is implied.

Exposure and session reporting. Supervision and Control prescribed two nominal five-minute task blocks per interface: one for Cup Insertion and one for T-Shape Assembly, following the assigned task order. Recorded block durations could be longer. Three interface conditions give 30 min of planned measured exposure per participant. Fleet used four 7.5-min conditions (30 min), following the same comprehension and successful-practice requirements as the interface studies. Complete sessions lasted approximately 90 min, including briefing, checks, practice, questionnaires, and breaks; this differs from the 30 min of prescribed measured interaction.

The runtime audit found no automatic 300-s timer or explicit measured-interaction start/stop markers. Launcher timestamps include child startup, preparation, Ready, correction, and shutdown; selected Control launcher lifetimes average 36.725 min per participant. The previously reported 37.4-min Supervision total could not be independently reproduced because of incomplete, restarted, or unusually long records and is not used as verified exposure. The experimenter-reported five-minute task periods remain the prescribed duration; their exact start/stop convention requires the experimenter script. Policy/evaluator clocks pause during takeover and correction in Supervision/Control, whereas the Fleet schedule clock continues. Fleet’s 450 s is a scheduling horizon, not an automatic process stop. Primary Fleet analyses retain their declared 450-s observation windows; administrative or launcher timestamps do not redefine these windows.

Software provenance. The Control report identifies Python preprocessing and references both `python_environment.json` and `R_sessionInfo.txt`. The Fleet README documents Python with pandas, NumPy, and SciPy, and R with `lme4`, `lmerTest`, `emmeans`, and `survival`, alongside data-processing and plotting packages. Its Python 3.10+ entry is a dependency requirement, not a verified executed version. The original per-study environment manifests remain incomplete. The common-recovery refits supply a separate R session log. Versions for the original analyses are not established by the available files. The local LaTeX/figure-generation environment does not establish the environment used to fit the original statistical models.

S12 Retrospective Outcome Harmonization and Protocol Audit

The common attempt-level endpoint was constructed retrospectively after collection. The original Supervision immediate-only endpoint and the Control endpoint were documented before this harmonization; neither is represented as prespecified. No cross-study model or contrast is estimated.

S12.1 Event attribution and denominators

An attempt succeeds at its first valid post-release check, or through same-cell, same-episode terminal success after documented autonomous resumption and before another intervention request. A later request before success or a terminal non-success makes that attempt unsuccessful. Missing terminal follow-up, an unexplained competing episode, or otherwise unascertainable outcome at the valid cutoff remains unresolved. A later correction’s success is never credited backward. Matching prefers an exact intervention identifier; where it is missing or stale, the first chronological same-episode check before the next request is used. No task-state event occurred during active human control in the audited reconstructed attempts.

Table S15: Common recovery accounting. All 24 participants in each cohort contribute outcomes; model units are attempts.

Study	Retained	Immediate	Continuation	Unsuccessful	Unresolved	Known
Supervision	708	407	34	190	77	631
Control	910	652	35	170	53	857

The Control unresolved counts are KT 10, MC 23, and FACTR 20. All end without a terminal record within valid follow-up, and no later logged outcome legitimately resolves them. All studies exclude active-control attempts lasting 1.5 s or less. The original Supervision and Control implementations used a strict less-than boundary, but no audited attempt lasted exactly 1.5 s; the unified wording therefore leaves the retained samples unchanged.

S12.2 Model and marginal results

Separate binomial-logit models use condition, task, their interaction, and condition order, with participant and repeated physical-scenario random intercepts. The scenario identifier combines task, settled scene-position hash, model variant, and OOD-scene hash. It is not an attempt identifier. Of 111 Supervision scenario levels, 82 recur across episodes and participants; of 126 Control levels, 99 recur across episodes and 98 across participants. Neither fit is singular. Marginal probabilities equally weight the two tasks. Confidence intervals are unadjusted 95% intervals; Holm correction covers the three within-study condition contrasts.

Table S16: Harmonized recovery probabilities, adjusted and equally averaged across tasks.

Study	Condition	recovery (%)	95% CI
Supervision	Desktop-RGB	68.4	[58.5, 76.8]
Supervision	VR-RGB	56.0	[45.8, 65.7]
Supervision	VR-PointCloud	83.6	[77.3, 88.4]
Control	KT	88.9	[83.0, 92.9]
Control	MC	78.3	[70.1, 84.7]
Control	FACTR	84.7	[77.2, 90.0]

Table S17: Harmonized recovery contrasts. OR divides the first condition by the second.

Study	Contrast	OR	95% CI	Holm p
Supervision	Desktop-RGB / VR-RGB	1.697	[1.041, 2.767]	= .034
Supervision	Desktop-RGB / VR-PointCloud	0.423	[0.265, 0.674]	< .001
Supervision	VR-RGB / VR-PointCloud	0.249	[0.154, 0.403]	< .001
Control	KT / MC	2.212	[1.393, 3.511]	= .002
Control	KT / FACTR	1.442	[0.870, 2.389]	= .156
Control	MC / FACTR	0.652	[0.415, 1.024]	= .127

S12.3 Sensitivity of the Control finding

The primary KT/MC contrast is OR 2.212, 95% CI [1.393, 3.511], $p_H = .002$. The original participant-summary Friedman test is omnibus across three controllers ($\chi^2(2) = 3.851$, $W = .080$, $p_H = .292$ across six omnibus outcome sensitivities). It is not a KT/MC pairwise test. A paired Wilcoxon KT/MC follow-up on participant recovery proportions has median paired difference .188 and $p_H = .055$ across the three controller pairs; its confidence interval is not supplied. Aggregation removes task/order/scenario adjustment and changes weighting.

Including known outcomes from short attempts gives KT/MC OR 1.635, 95% CI [1.074, 2.490], $p_H = .066$ (890 known attempts). Model fits assigning all unresolved Control attempts to failure give OR 2.232 [1.475, 3.377], $p_H < .001$; assigning all to success gives OR 2.004 [1.272, 3.159], $p_H = .008$. These uniform extreme assignments do not exhaust all condition-dependent missingness mechanisms. The primary recovery contrast is therefore not described as fully robust.

For Supervision, retaining short attempts gives Desktop-RGB/VR-RGB OR 1.792 [1.111, 2.889], $p_H = .017$; Desktop-RGB/VR-PointCloud OR .474 [.303, .740], $p_H = .002$; and VR-RGB/VR-PointCloud OR .264 [.168, .416], $p_H < .001$. The recovery contrasts retain their direction and survive correction under this sensitivity (656 known attempts).

S12.4 Scenario-bank construction and detector scope

The author-reported ID perturbation bounds are 5 cm per horizontal axis and 15° for T-block/box yaw; cup yaw is not randomized. The archived generator available for audit instead specifies 1 cm and 5° . The reported bounds and archived defaults are therefore distinguished here; the effective study settings require the corresponding launch configuration. For T-Shape OOD scenes, T1’s nominal position is (.600, $-.225$) m, with generated $x \in [.500, .700]$ m and $y \in [-.350, -.250]$ m. Its heading offsets lie approximately in $[-28.65, -20.05]^\circ$ or $[20.05, 57.30]^\circ$. T2’s nominal position is (.600, $-.025$) m, with $x \in [.550, .650]$ m and $y \in [.000, .030]$ m, and heading offsets approximately $[-19.48, -11.46]^\circ$ or $[11.46, 20.05]^\circ$. Positions and headings are changed jointly; there is no common radial/yaw interval shared by both objects.

Cup OOD scenes exchange the cracker- and sugar-box texture maps and independently change each cup’s height scaling from its nominal value to 75–85% or 110–125% of nominal, while retaining ID position/box-yaw perturbations. The hash audit matches the observed A/B OOD scene files to the local human-study banks. These settings are not a separate audit of Fleet or offline-learning banks. Frozen training manifests are still required to establish exact disjointness from policy-training support.

The collected-release source audit supersedes the earlier current-default detector description. The thresholds below are attributed from recorded releases; they were not serialized per launch, and all primary Fleet blocks record a dirty umbrella worktree. Exact per-launch overrides therefore remain unresolved. T-Shape completion requires grasping and lifting T1 by at least .08 m, bar contact, normalized face overlap at least .40, target-top alignment at least .75, face alignment at least .85, stem-up alignment at least .75, stem height at least .03 m, face-gap magnitude at most .025 m, no forbidden stem contact, and release opening at least .025. Supervision/Control release defaults specify distance $< .02$ m and bar-axis error at most 25° . Two early Fleet blocks map to these defaults, two to $< .04$ m/ 30° , and the final 92 to $< .04$ m/ 45° . The full-archive counts describe executed configurations, not the 95-block primary analysis denominator.

Both designated Cup pairs must satisfy radial error at most .040 m, vertical error within .025 m of a $+.0035$ -m offset, relative-axis error at most 20° , containment margin at least $-.005$ m, and pair contact. Cups and boxes must remain upright within 35° and 20° , respectively. Speed, gripper-opening, bad-contact, and wrong-cup gates were effectively disabled. These criteria were unchanged across the audited Cup releases.

The specialized task branches incremented a contact counter but did not enforce ten stable observations. Release checks therefore evaluate the corrected state once, before another policy action. Success ends the episode; otherwise autonomy resumes. All 7,614 automatic failures in the runtime audit occurred at policy-step cutoffs (127 for T-Shape, 217 for Cups); no wall-clock timeouts occurred. Learned-policy commands were paused during human correction, with controller assistance/filtering but no learned-policy blending. These archive-wide event totals are not recovery-model denominators.

S12.5 Delivered scenarios and protocol reconciliation

The runtime audit reconstructs active OOD membership from episode-start flags until the cell’s next episode. Across 278 bounded Supervision/Control blocks, OOD scenes account for 4,725.861 of 14,851.506 observed robot-minutes (31.82%). Time-weighted active OOD counts range from 2.730 to 2.924 across interface/task groups; counts are two or three essentially throughout, apart from a momentary reconstructed count of four in Control/MC/T-Shape. Episode-start

proportions and time-weighted exposure have different denominators. Operator selection and intervention requests are not direct annotations of assistance need in these two studies.

Fleet targeted 0.4 scheduled incidents per robot-minute. Across all 96 archived blocks, 1,395 incidents were scheduled, 1,392 eventually activated, and 1,386 activated within logged condition boundaries. Activation waits for a safe episode boundary; scheduling, activation, and effective need are distinct events. The primary analysis instead uses 95 eligible 450-s windows: F1/F3/F6/F9 contain 70/220/441/661 scheduled incidents, 70/211/432/644 activations in those windows, and 162/372/854/1,275 effective needs. Expected-demand normalization uses the target rate of 0.4 incidents per robot-minute.

Scheduled grasp/lift-failure labels total 33/37, 110/110, 168/273, and 168/493 for F1/F3/F6/F9 in the primary sample. The increasing lift-failure share means that difficulty-label composition was not balanced across fleet sizes. These labels are not a validated severity scale. Section S13 reports reconciled useful-throughput sensitivity fits under archived configuration exclusions, together with their interpretation limits. Current primary estimates and exclusions are unchanged. Accordingly, the Fleet comparison cannot isolate robot count from realized demand composition.

S12.6 Missingness and task dependence

All 130 unresolved Supervision/Control attempts reached the valid observation boundary without an attributable outcome. Six Supervision records contain terminal events only after that cutoff (five failures and one success, 3.45–99.52 s later); these remain unresolved under the declared rule. Missingness differs by condition and task, so uniform all-success/all-failure assignments do not bound all differential missingness patterns.

Control task-specific recovery is .923/.746/.836 for KT/MC/FACTR in Cup Insertion and .842/.816/.857 in T-Shape Assembly. Exploratory reference-coded condition-by-task coefficients were OR 1.284 ($p = .615$) for Supervision VR-RGB and .197 ($p < .001$) for VR-PointCloud, relative to Desktop-RGB; Control coefficients were 3.397 ($p = .009$) for MC and 2.637 ($p = .056$) for FACTR, relative to KT. These raw coefficient tests are not a multiplicity-adjusted global interaction test. Section S14 reports all harmonized Supervision task-specific marginal estimates and contrasts from the reproduced fit. Task-averaged findings should not be read as uniform effects across tasks.

Supervision contains 144 authoritative participant-interface-task blocks. Two distinct VR-RGB/T-Shape blocks carried the same erroneous embedded participant identifier; deduplicating that field produced the earlier count of 143. Cohort-folder identifiers distinguish both blocks, which remain included in the unchanged 631-known-outcome model. Control retains 143 task blocks after the documented unavailable FACTR/T-Shape cell. No missing block was imputed or relabeled; all 24 participants per study contribute outcomes.

S13 Reconciled Fleet Eligibility and Throughput Sensitivity

The fixed-window primary analysis retains 95 of 96 canonical blocks; the single 442.96-s F1 block remains sensitivity-only. Raw-event reconstruction reproduces all 2,663 effective needs and 756 intercepted needs, with no event-membership differences or failed temporal joins. The 892 requests and 882 control starts precede phase, horizon, and duration exclusions. Of 881 start-bounded correction intervals, 61 ended after the 450-s window and 48 lasted at most 1.5 s; these exclusions do not overlap and leave 772 eligible active corrections. Of 737 evaluable first post-release checks, 694 indicate success. Useful intervention throughput uses these successful corrections, not all autonomous episode successes.

The earlier reconstruction’s 1,541 successful episodes remains a separate descriptive outcome. Its 881 correction intervals likewise used different eligibility. The reconciliation retains both definitions explicitly; neither replaces the primary outcome. Timing summaries distinguish pooled event means from equal-weight averages of block means. Primary useful-throughput estimates, eligibility, historical success labels, and Main Figure 7 are unchanged.

Exclusion sets and models. Omitting the two earliest-release blocks leaves 93 blocks from 24 participants. Omitting the participant whose four blocks used pre-final releases leaves 91 blocks from 23 participants. The archived configuration restrictions select these same 91 blocks and share a single fit. The omitted participant contributed F1/F3/F6/F9 at positions 1/2/3/4; the previously excluded short F1 block was at position 3. Archived configuration, participant, date, and order sequence remain inseparable, so the exclusions assess sensitivity without identifying the cause of any difference.

Useful intervention counts use categorical fleet size, centered condition position, log condition-minutes as an offset, and participant random effects. The archived rule first fits a random intercept and linear fleet-size slope, simplifying for singularity or convergence at tolerance 10^{-4} . All three maximal candidates were singular; all participant-intercept Poisson fallbacks converged and were nonsingular. Pearson dispersion was .376, .324, and .321, respectively, below the 1.20 threshold for negative-binomial escalation. No final fit failed, and no model was selected by significance.

Table S18: Useful-intervention-throughput sensitivity. Joint Wald tests compare the three categorical fleet terms. None detects a fleet effect; this does not establish equal throughput.

Analysis	Blocks	Participants	$\chi^2(3)$	p
Reference	95	24	7.217	.065
Earliest-release omission	93	24	5.310	.150
Pre-final participant omission	91	23	4.984	.173

All six pairwise contrasts within each unique fit form one Holm family. Intervals are unadjusted 95% asymptotic Wald intervals on the log scale, back-transformed to rate ratios. Every Holm-adjusted comparison remains inconclusive (Table S19). These new exclusion refits concern useful throughput; they do not establish inferential robustness of coverage, delay, or correction-time effects under those same exclusions. Previously reported descriptive coverage stability and the full-archive model-reproduction checks answer separate questions.

Table S19: All useful-throughput sensitivity contrasts. Ratios divide the higher fleet size by the lower. Each six-contrast family is corrected separately.

Blocks	Contrast	Rate ratio	95% CI	Raw p	p_H
95	F3/F1	1.185	[0.949, 1.481]	.134	.538
95	F6/F1	1.285	[1.033, 1.599]	.024	.122
95	F9/F1	1.316	[1.059, 1.636]	.013	.080
95	F6/F3	1.084	[0.882, 1.333]	.441	.950
95	F9/F3	1.111	[0.904, 1.364]	.317	.950
95	F9/F6	1.024	[0.838, 1.252]	.817	.950
93	F3/F1	1.167	[0.933, 1.460]	.177	.709
93	F6/F1	1.240	[0.996, 1.545]	.054	.270
93	F9/F1	1.269	[1.020, 1.579]	.032	.194
93	F6/F3	1.063	[0.864, 1.309]	.564	1.000
93	F9/F3	1.088	[0.884, 1.338]	.425	1.000
93	F9/F6	1.023	[0.837, 1.251]	.823	1.000
91	F3/F1	1.167	[0.933, 1.460]	.177	.707
91	F6/F1	1.230	[0.986, 1.535]	.067	.334

Blocks	Contrast	Rate ratio	95% CI	Raw p	p_H
91	F9/F1	1.265	[1.014, 1.577]	.037	.222
91	F6/F3	1.054	[0.854, 1.301]	.623	1.000
91	F9/F3	1.084	[0.879, 1.336]	.451	1.000
91	F9/F6	1.028	[0.836, 1.264]	.793	1.000

The archived primary sensitivities, including full logged intervals, inclusion of the short block, omission of order, and leave-one-participant-out fits, were also reproduced in the full Fleet audit. The separate participant-summary throughput Friedman omnibus remained significant ($\chi^2(3) = 13.259$, $p = .004$), unlike the primary mixed-model omnibus; this aggregation-dependent result does not replace the primary adjusted contrasts. Cross-package checks against the manuscript companion match the 95/93/91 block sets and all six primary throughput contrasts. No additional companion-only failed-fit record was identified.

S14 Exploratory Supervision recovery by Task

The reproduced harmonized interaction model uses 631 known recovery outcomes, participant and physical-scenario random intercepts, interface, task, their interaction, and interface order. It converged and was nonsingular at tolerance 10^{-4} . The task-specific estimates below average recorded interface-order levels equally and use asymptotic Wald intervals on the logit scale, back-transformed to probabilities. They use the common recovery endpoint, not the earlier immediate-only endpoint.

Table S20: Exploratory task-specific Supervision recovery. Intervals are unadjusted 95% CIs; percentages are model estimates.

Task	Interface	recovery (%)	95% CI
Cup Insertion	Desktop-RGB	55.8	[41.3, 69.4]
Cup Insertion	VR-RGB	39.7	[26.7, 54.3]
Cup Insertion	VR-PointCloud	87.1	[78.6, 92.5]
T-Shape Assembly	Desktop-RGB	78.7	[67.6, 86.7]
T-Shape Assembly	VR-RGB	71.1	[59.0, 80.9]
T-Shape Assembly	VR-PointCloud	79.5	[70.9, 86.0]

VR-PointCloud exceeds both RGB interfaces for Cup Insertion under the six-comparison Holm family. None of the T-Shape comparisons survives correction; this does not establish equivalence. Thus the task-averaged advantage should not be interpreted as an equally supported advantage on both tasks. These follow-ups are exploratory and do not isolate task mechanics from the associated scene distributions.

Table S21: All exploratory Supervision task-specific recovery contrasts. Odds ratios divide the first interface by the second; Holm correction covers all six comparisons together.

Task	Contrast	OR	95% CI	Raw p	p_H
Cups	Desktop / VR-RGB	1.923	[0.955, 3.874]	.067	.269
Cups	Desktop / PointCloud	0.188	[0.093, 0.379]	< .001	< .001
Cups	VR-RGB / PointCloud	0.098	[0.047, 0.202]	< .001	< .001
T-Shape	Desktop / VR-RGB	1.497	[0.758, 2.957]	.245	.490
T-Shape	Desktop / PointCloud	0.954	[0.507, 1.793]	.883	.883
T-Shape	VR-RGB / PointCloud	0.637	[0.342, 1.187]	.155	.466

S15 Successful Interventions per Recorded Block Minute

This additional, review-requested analysis complements the attempt-level recovery models. It retains the same success definition and attempt-eligibility criteria while counting only successes documented within each block’s validated observation window. The counts include 34 Supervision and 35 Control successes during autonomous continuation. One Supervision outcome first documented after cutoff remains unresolved for this analysis. Unresolved outcomes do not contribute successful events; they are not recoded as observed failures. Rates use the eligible block exposure rather than the number of ascertained binary outcomes.

The rate dataset contains 144 Supervision and 143 Control task blocks, with 24 participants in each independent cohort. The unavailable Control FACTR–T-Shape Assembly block was not imputed. For 277 of the 287 observed blocks, exposure is the difference between the monotonic block-start and block-end timestamps. The launcher captures the start before policy and interface startup. The end generally follows child-process cleanup. Practice occurred before these selected blocks and is excluded.

Ten Supervision blocks required event-derived cutoffs because their metadata ends were missing or implausible. Nine ended at the last non-BLOCK_END behavior event within 600 s, and one ended at the last participant-protocol event. Their exposure runs from the recorded block-start timestamp to the validated event cutoff. Later events contribute neither additional exposure nor successful outcomes.

Total recorded exposure was 870.877 min in Supervision and 881.401 min in Control, compared with 720 and 715 nominal task-block minutes, respectively. The launchers did not automatically terminate these task blocks at 300 s. Startup and shutdown explain only part of the difference: 283 of 287 blocks contained more than 300 s between their earliest episode start and last included behavior event. The logs cannot establish whether individual overruns reflected completion of an active episode or intervention, delayed stopping, or another procedural convention. Accordingly, the denominator is recorded block wall-clock exposure, not an exactly fixed interaction interval or the separate episode clock that pauses during intervention.

Table S22 distinguishes pooled observed rates from model-estimated rates. The two Poisson mixed models converged and were nonsingular; reported dispersion statistics were 0.596 for Supervision and 0.493 for Control. Negative-binomial escalation was not triggered. The three condition contrasts were Holm-corrected separately within each study. All three Supervision comparisons survived correction. Complete Control contrasts are given in Table S23. The existing primary analyses were unchanged.

Table S22: Successful interventions per recorded block minute. Exposure totals by condition are rounded to one decimal. Pooled rates divide observed successes by exposure; model estimates and their 95% confidence intervals come from the separate Poisson mixed models.

Study	Interface	Successes	Minutes	Pooled rate	Model rate [95% CI]
Supervision	Desktop-RGB	115	276.6	0.416	0.400 [0.330, 0.486]
	VR-RGB	89	294.1	0.303	0.293 [0.236, 0.364]
	VR-PointCloud	237	300.2	0.790	0.766 [0.665, 0.882]
Control	KT	253	294.8	0.858	0.805 [0.666, 0.973]
	MC	245	295.3	0.830	0.751 [0.619, 0.911]
	FACTR	189	291.2	0.649	0.602 [0.491, 0.737]

Table S23: Control rate ratios for successful interventions per recorded block minute. Ratios follow the listed numerator/denominator direction. Confidence intervals are 95%; p_H is Holm-corrected across these three comparisons.

Contrast	Rate ratio	95% CI	p_H
KT / MC	1.072	[0.899, 1.280]	.438
KT / FACTR	1.338	[1.109, 1.616]	.007
MC / FACTR	1.248	[1.032, 1.509]	.045

S16 System Performance and Visual Update Rates

System characterization distinguishes the rate at which new scene information reaches an operator from the rate at which the display redraws. This distinction matters for evaluating the implemented supervision interfaces.

Measurement scope. Except for point-cloud publishing and intervention-loop rates extracted from study logs, measurements were collected on 17–18 September 2026 in dedicated 1.5–3-min runs on the study machine, launcher, and reported configuration. The topology comprised nine cells with one selected for detailed inspection. Values generally summarize medians of 10-s windows. Headset display rates were recorded without the latency probe; latency runs included it. These measurements characterize the reproduced implementation and do not establish per-participant display performance.

Table S24: Essential visual-performance measures. Scene refresh and display frame rate refer to different stages. Except for the in-study point-cloud publish rate, these are post-study measurements. Latencies are not photon-level measurements.

Interface	New scene information	Display (fps)	State-to-presentation latency
Desktop-RGB	60.0-fps local rendering	60	48.9–49.5 ms; excludes monitor input lag
VR-RGB	Approximately 59-Hz reception; image decoding capped at 10 Hz	68–70	52.9–55.5 ms; measured upper-bound proxy
VR-PointCloud	28.5-Hz publishing per camera in-study; 58.9–60.2-Hz cloud redraw in later runs	60–61	Approximately 31–62 ms; component estimate

Configuration and control. MuJoCo used a 1-ms timestep. The autonomous policy ran at 10 Hz, and the interface state mirror was configured at 30 Hz. Across recorded episodes containing human control, achieved intervention-loop rates were 54.3–55.1 Hz against a 60-Hz target; the FACTR leader hardware loop was configured at 500 Hz. These are separate loops. Desktop-RGB used a 1600×900 nine-cell grid with vertical synchronization. Quest 3 interfaces used Wi-Fi and a 72-Hz display target. VR-RGB used 480×360 JPEG images at quality 75 for wrist, right, and left panels plus an overview. VR-PointCloud used top, right, and left 640×480 depth views, pixel stride 2, and a 30-Hz per-camera publish cap. Median combined point counts were approximately 25,900 for T-Shape and 37,600 for Cups.

Temporal fidelity and interpretation. Post-study measurements confirmed that new VR-RGB image updates were slower than stream reception. The detailed camera panels reused an overview component with a 10-Hz decode limit; the reception and display rates in Table S24 do not represent new-image refresh. The report identifies this limit as predating the study with no overriding build setting; its presence during participation is therefore inferred rather than directly measured. The newest available frame was displayed, so low image refresh can coexist with relatively low latency for displayed frames. Point-cloud publishing measured 28.5 Hz per camera across 2,063 in-study single-view visits (5th–95th percentiles 25.2–29.3 Hz); later runs reproduced that rate within 1%. Point-cloud rendering used approximately 89–91% headset GPU capacity and reached 60–61 fps rather than the 72-Hz target. These implementation differences prevent attributing Supervision effects solely to spatial representation or immersion.

Latency boundaries. The RGB probe stamped frames and measured PC-clock round trips to decode and presentation acknowledgements. A presentation acknowledgement includes return transport, so it supplies an upper bound for the forward path rather than a synchronized one-way or photon-level delay. The reported state-to-presentation summaries combine component measurements. Desktop values omit monitor input lag. Point-cloud latency was not directly stamped; its 31–62-ms range sums measured and estimated components, including a network allowance derived from smaller RGB packets. It is not a strict bound. The probe reduced VR-RGB display rate by approximately 6–8 fps, so probe-on latency and probe-off display rate describe separate runs. Stage-wise percentiles were not added.

The accompanying `system_assets/SYSTEM_METRICS_REPORT.md` preserves the source report and its log/tool identifiers. The report summarizes system characterization; it does not establish per-participant latency.